CS486C – Senior Capstone Design in Computer Science
Project Description

Project Title: JacksRide: A Secure University Ridesharing Platform for Sustainable Long-Distance Transportation	
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Project Overview:
Transportation is one of the largest sources of greenhouse gas emissions in the United States and remains a significant challenge for universities seeking to reduce their environmental footprint. At Northern Arizona University (NAU), many students bring personal vehicles primarily because they need transportation between campus and their homes during holidays, long weekends, and academic breaks. Survey results conducted with NAU students in Spring 2026 indicate that more than half of students drive alone when traveling long distances, while many others depend on informal carpooling arrangements. Although NAU has successfully promoted sustainable commuting through its "Ditch the Car/No Car-No Problem" initiative, existing efforts focus primarily on local commuter travel rather than long-distance transportation needs. As a result, students continue to rely heavily on personal vehicles, increasing transportation costs, parking demand, roadway congestion, and greenhouse gas emissions.

The Office of Sustainability at Northern Arizona University works to reduce environmental impacts, improve campus sustainability, and support the university's commitment to carbon neutrality. Through ongoing transportation planning efforts, student surveys, and stakeholder engagement activities with departments including NAUgo, University Transit Services, and the School of Informatics, Computing, and Cyber Systems (SICCS), a significant opportunity has been identified to create a secure ridesharing platform specifically designed for the NAU community.
Currently, students seeking rides can post requests on social media groups, messaging platforms, or informal networks. These approaches have several shortcomings:

- No centralized platform exists for coordinating rides.
- Social media arrangements provide limited accountability and verification.
- Students have concerns regarding safety and reliability.
- Existing third-party rideshare applications do not verify university affiliation and prepackaged soltuions are expensive.
- Students often struggle to find transportation options that match their destinations, budgets, and schedules.
- Transportation costs can create economic barriers for students, particularly those from lower-income households where university affordability is an overriding concern.

To address these challenges, we envision JacksRide **(figure1)**, a secure web-based ridesharing platform available exclusively to verified members of the NAU community. The system will help students, faculty, and staff coordinate rides for both local and long-distance travel while improving safety, reducing transportation costs, and supporting sustainability goals. Users will authenticate through NAU's Single Sign-On (SSO) system and create profiles as either riders or drivers. The platform will allow users to create ride offers or ride requests, search for matching destinations, communicate through built-in messaging tools, and establish trusted transportation connections

within the university community.

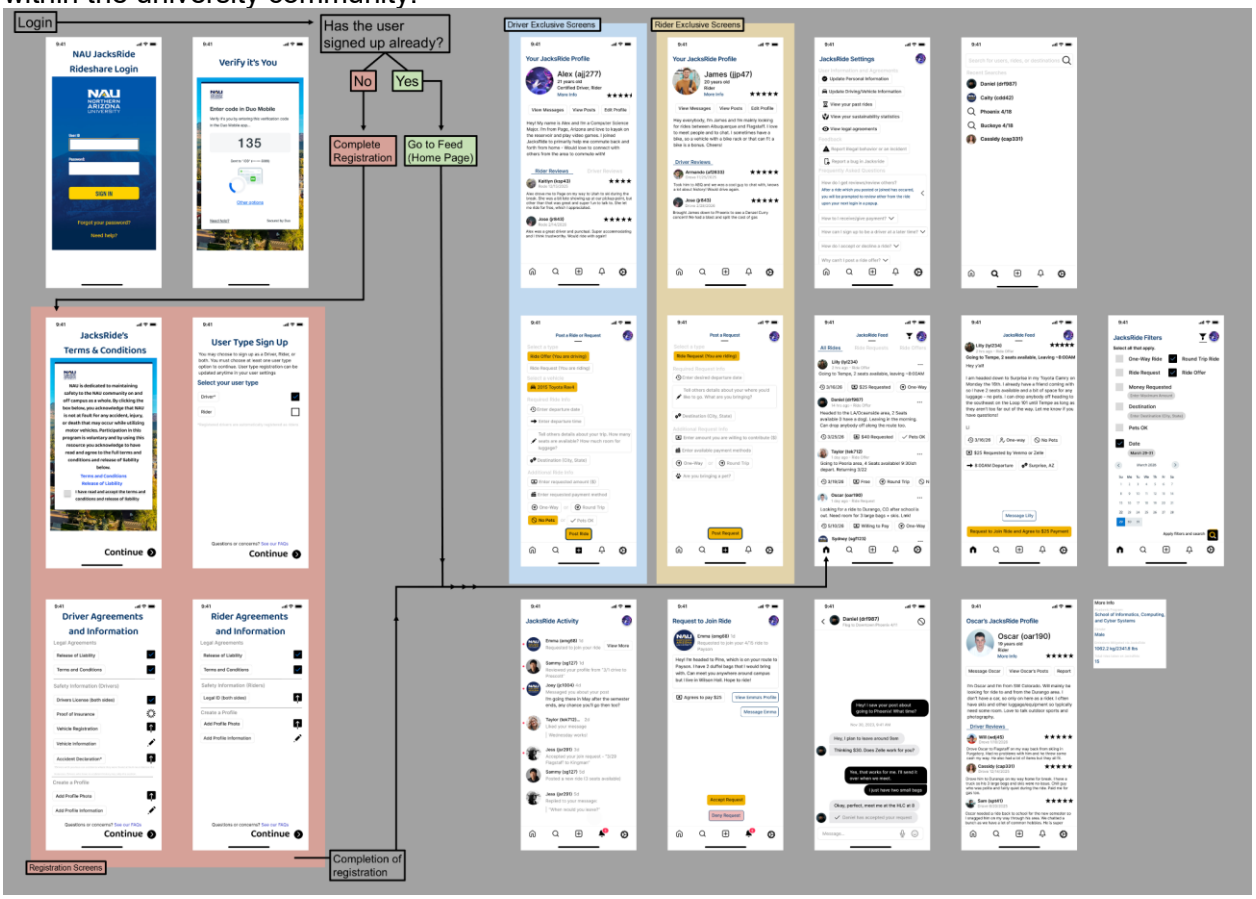


Figure 1: Wireframe outlining potential design of Jacks ride

The proposed solution would be developed as a modern web application, or NAUgo page, with a mobile-responsive interface. The system would include secure user authentication, role-based account management, ride posting capabilities, search functionality, messaging tools, administrative oversight functions, and data management features.

A successful JacksRide platform would provide a meaningful service for the NAU community while offering substantial educational value to the student development team. Thousands of students could benefit from lower transportation costs, safer ride coordination, and increased access to sustainable transportation alternatives. The platform would support NAU's Climate Action Plan, reduce reliance on single-occupancy vehicles, strengthen community connections, and provide a scalable solution that could potentially be adapted by other universities facing similar transportation challenges. The completed project would have immediate practical value and a realistic pathway toward implementation and deployment at Northern Arizona University.

Students will gain hands-on experience designing, building, testing, and deploying a real-world software system with an identifiable customer base and measurable sustainability impacts.

Knowledge, skills, and expertise required for this project:

- Full-stack web application development
- Database design and management
- REST API development
- User authentication and authorization
- Secure software engineering principles

- UI/UX design
- Mobile-responsive web development
- Cloud deployment and hosting
- Software testing and quality assurance
- Agile software development methodologies

Equipment Requirements:

- There should be no equipment or software required other than a development platform and software/tools freely available online.

Software and other Deliverables:

- A strong as-built report detailing the design and implementation of the product in a complete, clear and professional manner. This document should provide a strong basis for future development of the product.
- Complete professionally-documented codebase, delivered both as a repository in GitHub, BitBucket, or some other version control repository; and as a physical archive on a USB drive.
- Functional JacksRide web application.
- User interface designs and wireframes.
- Administrative dashboard and moderation tools.
- Authentication and user management system.
- Potentially: User guide and onboarding documentation.

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